Assignment - 04

CN

(22BCON1068)

• Section-A (2x5=10)

Q1. What is TCP ?

Ans:) TCP stands for Transmission control protocol. it is the main protocol of the internet protocol suite. it lies between the application and network layer which are used in providing reliable delievery services. it is connection oriented protocol for communication that helps in exchange of message between different devices over a network.

Q2. Define UDP ?

Ans UDP stands for User Datagram Protocol. The UDP allows the computer applications to send the message in the form of datagrams from one machine to another machine over the Internet protocol (IP) network. The UDP is an alternative communication protocol to the TCP. like TCP, UDP provides a set of rules that governs how the data should be exchanged over the Internet.

Q3. Define the term congestion control.

Ans Congestion control is a mechanism that controls the entry of data packets into the Network, enabling a better use of a shared network infrastructure and avoiding congestion collapse. Congestive - Avoidence Algorithm (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network.

Q4. What is Intra Domain routing.

Ans Indradomain Routing is the routing protocol that operates only within a domain. in other words, intradomain routing protocols are used to route packets within a specific domain, such as within an institutional network for e-mail or web browsing. it doesnot communicative with other domains.

Q5. Define MAC.

Ans MAC stands for "Media Access Control". it is sub-layer of the data link layer in the OSI model. The MAC address is a unique hardware address assigned to each network interface card (NIC) in a device, such as computer or router. This address is used to identify devices on a local network and is essential for data link layer operations, including Ethernet communication.

• Section-B (3x7 = 21)

Q1. What is transport layer? Explain reliable and unreliable in transport layer.

Ans. Transport layer : it is the layer 4 of the OSI model. it manages network traffic between hosts and end systems to ensure complete data transfers. Transport layer protocol such as TCP, UDP, DCCP and SCTP are used to control the volume of data, where it is sent, and at what rate. The transport layer uses reliable and unreliable connections to transfer data between two devices in a network.

• Reliable layer :- Reliable connection uses TCP (Transport Control Protocol). The TCP protocol uses a three way handshake, windowing and sequence numbers for guranteed data delivery.

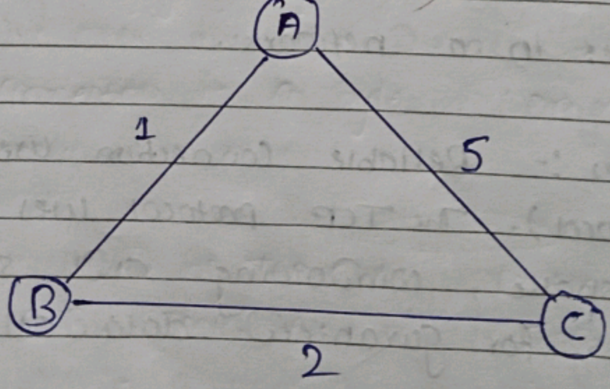
Unreliable : unreliable transport Protocols, such as UDP (user Datagram Protocols), do not guarentee reliable data delivery. They Provide minimal error checking and correction, and they do not establish a connection before transmitting data.

Q2. Explain ~~efficiency~~ distance vector routing with diagram.

Ans A ~~data~~ distance vector routing Protocol in data networks determines the best route for data Packets based on distance. Distance vector routing protocols measure the distance by the number of routers a packet has to pass; one router counts as one hop. Distance-vector routing protocols also require that a router inform its neighbours of network topology changes periodically. Historically known as the old ARPANET routing algorithm (or known as ~~the Old~~ Bellman-ford Algorithm).

The distance vector routing algorithm is iterative asynchronous and distributed. it is a dynamic algorithm and mainly used in ARPANET and RIP (Routing Information Protocol). Each router maintains a distance table known as vector consider the following network: three routers are three :

A, B and C with AB = 1, BC = 2, and CA = 5



Q8. Explain efficiency considerations in detail.

Ans. Efficiency considerations in computer networks are essential for optimizing performance, minimizing resource usage and ensuring a reliable and responsive communication infrastructure. Here are key aspects of efficiency in computer Network:

(i) Bandwidth utilization : Efficient Network protocols, such as transmission control protocol (TCP) and user Datagram protocol (UDP), help in minimizing overhead and maximizing data transfer rates.

(ii) Latency and Delay : Employing efficient routing algorithms to determine the shortest paths and reduce packet traversal times.

(iii) Reliability and Redundancy : Creating redundant paths to optimize and minimize the impact of network failure and enhance reliability.

(iv) Resource utilization : Deploying efficient network devices and hardware to handle increased traffic without performance degradation.

(v) Security Consideration :- Efficiently filtering and inspecting network traffic to identify and mitigate potential security threats.

(vi) Scalability : Designing network architecture that can scale horizontally or vertically to accomodate increasing numbers of users and devices.

• Section - C (3x11=33)

Q1 Differentiate between TCP and UDP protocol.

Ans

→	TCP	UDP
1.	Full form : Transmission Control Protocol	Full form : User Datagram Protocol
2.	Requires an established connection before transmitting the data.	No connection is needed to start and end a data transfer.
3.	Can sequence data (send in specific order).	Cannot sequence or arrange data.
4.	Can Retransmit data if packet fail to arrive.	No data retransmitting. if data can't be received.
5.	Delivery is guarented.	Delivery is not guranted.
6.	Broadcasting is not supported.	Broadcasting is supported in UDP.
7.	Acknowledgement segment is present.	Acknowledgement segment is not present.
8.	Speed is comparatively slower than UDP.	UDP is faster, simpler and more efficient than TCP.
9.	TCP is heavy weight.	UDP is light weight.
10.	TCP connection is a byte stream.	UDP connection is a message stream.
11.	Protocols used by TCP are HTTP, FTP, STMP, etc.	Protocols used by UDP are RIP, TFTP, DNS, etc.

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Q. Describe wireless network 802.11 in MAC.

Ans The 802.11 standard, commonly known as wi-fi, comprises both the physical layer and the Medium Access control (MAC) layer. MAC layer provides functionally for several tasks like control medium access, can also offer support for roaming, authentication and power conservation. The basic services provided by MAC are the mandatory asynchronous data service and optimal time-Bounded service. IEEE 802.11 defines two MAC sub layers :-

1. Distributed Coordination function (DCF) : DCF uses CSMA/CA as access method as wireless LAN can't implement CSMA/CD. it only offers asynchronous service.
2. Point Coordination function (PCF) : PCF is implemented on top of DCF and mostly used for time-service transmission.

MAC frame : The MAC layer frame consists of 9 fields. The following figure shows basic structure of an IEEE 802.11 MAC data frame.

FRAME CONTROL	DURATION /ID	Address 1	Address 2	Address 3	SC	Address 4	Data	CRC
2 Bytes	2 Bytes	6 Bytes	6 Byte	6 Bytes	2 Bytes	6 Byte	0-2312 Bytes	4 Bytes

- Frame control - it is 2 Bytes long field which defines type of frame and some control information.
- Duration/ID - it is a 4 bytes long field which contains the value indicating the period of time in which the medium is occupied.
- Address 1 to 4 - There are 6 bytes long fields which contain standard IEEE 802 MAC addresses. (48 bit each).
- SC (Sequence Control) - it is a 16 bits (2 bytes) long field

which consists of two sub-fields i.e; seauence number (12 bits) and fragment number (4 bits).

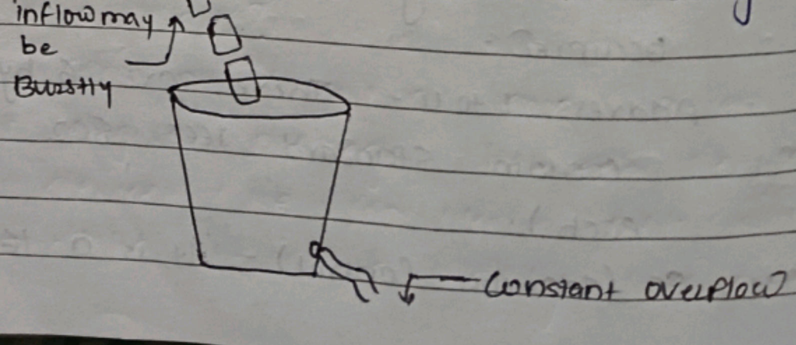
- Data - it is a variable length field carries the data from upper layers.
- CRC (cyclic redundancy check) - it is a 4 bytes long field which contains a 32 bit CRC error detection seauence to ensure error free frame.

Q.3. Explain congenction control algorithms.

Ans Congestion control Algorithm : Congestion control is a mechanism that controls the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.

There are two congestion control algorithm :

(i) Leaky Bucket Algorithm : The leaky bucket algorithm discovers its use in the context of network traffic shaking or rate limiting. A leaky bucket algorithm and a token bucket execution are predominetly used for traffic shacking algorithms. This algorithm is used to control the rate at which traffic is sent to the Network and shape the burst traffic to a steady traffic stream. This disadvantages compared with the leaky-bucket algorithm are the inefficient use of available network resources. The Large area of network resources such as bandwidth is not being used efficitively.



(ii) Token Bucket Algorithm : The leaky bucket algorithm has a rigid output design at an average rate independent of the burstty traffic. in some applications, when large bursts arrive, the output is allowed to speed up. This calls for a more flexible algorithm, puferably one that never loses information. Therefore a token Bucket algorithm finds its uses in network traffic shaping or rate-limiting. it is a control algorithm that indicates when traffic should be sent. This order comes based on the display of tokens in the bucket. The bucket contains tokens. Each of the token defines a packet of predetermined size. Tokens in the bucket are deleted for the ability to share a packet. When tokens are shown, a flow to transmit traffic appears in the display of tokens. No Token means no flow sends its packets. Hence, a flow transfers traffic up to its peak burst rate in good token's in the bucket.

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